

WILLIAM JACKSON

Senior Software Engineer – Node (Express/NestJS) AWS/Azure

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PROFESSIONAL SUMMARY

Senior Software Engineer with 10+ years of experience building **backend platforms**, **API-first systems**, and **cloud-native services** for networking, business workflow, and SaaS-driven product environments. Strong background designing resilient services with **Node.js 14–22**, **TypeScript 4.x–5.x**, **Express.js 4.x–5.x**, and **NestJS 9.x–10.x**, with practical ownership across service decomposition, event-driven processing, relational data design, caching, and production reliability. Delivers realistic engineering value by solving latency issues, stabilizing error-prone integrations, modernizing legacy code paths, and implementing secure backend capabilities that improve maintainability, release confidence, and operational visibility.

SKILLS

- **Backend:** **Node.js 14–22**, **TypeScript 4.x–5.x**, **JavaScript ES6+**, **Express.js 4.x–5.x**, **NestJS 9.x–10.x**, REST API design, microservices, backend modularization, server-side validation, authentication flows, authorization design, background job processing, event-driven architecture, API integrations
- **Data & Messaging:** **PostgreSQL**, **MySQL**, **SQL Server**, **MongoDB**, **Redis**, schema design, query tuning, indexing, transaction handling, pagination strategy, caching, **Kafka**, **RabbitMQ**, **AWS SQS**, **AWS SNS**, asynchronous processing, retry handling, idempotent consumers
- **Cloud & DevOps:** **AWS**, **EC2**, **ECS**, **EKS**, **Lambda**, **API Gateway**, **S3**, **RDS**, **CloudFront**, **DynamoDB**, **ElastiCache**, **EventBridge**, **Step Functions**, **Docker**, **Kubernetes**, **Terraform**, **GitHub Actions**, **GitLab CI/CD**, release automation, environment configuration, containerized deployment
- **Observability / Quality / Security:** **OpenTelemetry**, **Datadog**, **Prometheus**, **Grafana**, **CloudWatch**, structured logging, metrics instrumentation, error diagnostics, health checks, integration testing, API contract validation, **JWT**, **OAuth2**, **RBAC**, audit-aware backend design, rate limiting, secure secret handling

WORK HISTORY

Senior Software Engineer

February 2021 to December 2025

Viaro Networks Inc. - Delaware, United States

- Architected and delivered **Node.js 18–22** backend services that supported configuration-heavy platform workflows, partner integrations, and internal operational tooling, improving service consistency while making complex business rules easier to maintain across multiple environments.
- Led migration of older server modules into a cleaner **TypeScript 4.x–5.x** service structure with reusable request handling patterns and safer validation layers, reducing regression-prone code duplication by **31%** during feature expansion and release preparation.
- Designed secure **REST APIs** with **Express.js 4.x–5.x** and **NestJS 9.x–10.x** for administration, reporting, and workflow orchestration, applying **JWT**-based authentication and **RBAC** enforcement that reduced manual access-review effort by **27%**.
- Solved a recurring production defect where large filtered result sets caused unstable response times by analyzing slow database access paths, adding targeted indexes, and rewriting pagination behavior, which improved high-volume endpoint response stability by **42%**.
- Implemented asynchronous processing with **AWS SQS** and event-driven workers for notification, synchronization, and downstream update tasks, reducing user-facing wait time across affected flows by **35%** and preventing fragile synchronous chains from blocking requests.
- Stabilized a difficult integration issue caused by partial third-party payload failures by adding defensive parsing, retry-aware processing, and structured failure logging, which lowered repeated support escalations by **24%** and improved traceability for external-system errors.
- Introduced **Redis 6.x–7.x** caching for frequently requested configuration and lookup data, cutting unnecessary database pressure while preserving correctness through scoped invalidation strategies aligned with backend update events.
- Built containerized deployment pipelines using **Docker 23.x–26.x**, **AWS**, and CI/CD automation so backend releases became more repeatable, easier to roll back, and less dependent on environment-specific manual setup across staging and production.
- Improved API reliability by standardizing validation, error serialization, and status-code behavior across services, giving frontend and integration consumers more predictable contracts and reducing time spent diagnosing inconsistent backend responses.
- Addressed a memory-heavy export problem that intermittently failed for large operational reports by replacing buffer-oriented generation logic with streamed processing, allowing the system to complete demanding export jobs without exhausting runtime resources.
- Added **OpenTelemetry 1.x**, structured logging, and service-level metrics around critical request paths so that latency spikes, downstream dependency issues, and recurring failure patterns could be identified much faster during production investigations.

- Collaborated with product and platform stakeholders to implement new backend capabilities in a realistic delivery sequence, balancing feature demand, system reliability, and incremental modernization so the platform could evolve without destabilizing active customer workflows.

Software Engineer
Avantia, Inc. - Ohio, United States

October 2018 to January 2021

- Built and maintained **Node.js 14–18** backend modules for internal business applications, focusing on service endpoints, database-backed workflow logic, and partner-facing integrations that required stable request handling and careful operational validation.
- Expanded backend functionality with **Express.js 4.x** APIs that supported user management, data processing, and reporting workflows, making older server behavior more modular and easier to enhance while improving backend delivery efficiency by **26%**.
- Resolved a recurring issue where concurrent updates produced inconsistent record state by tightening transaction boundaries, improving optimistic update handling, and refining validation order, which reduced multi-user data conflicts by **33%**.
- Improved query-heavy modules in **PostgreSQL 12–14** and **SQL Server 2016–2019** by analyzing execution patterns, reducing unnecessary joins, and aligning indexes with real filter usage, which improved response time for high-traffic operational screens by **38%**.
- Implemented background processing for time-consuming tasks such as imports, notifications, and status synchronization, reducing timeout risk in synchronous request paths and giving users faster feedback for long-running backend operations.
- Helped modernize legacy JavaScript services toward **TypeScript 4.x**, introducing stronger typing around DTOs, service interfaces, and shared utility code, which reduced avoidable runtime bugs by **22%** and improved confidence during review and testing.
- Investigated and fixed an intermittent integration bug caused by malformed upstream data, then added stricter schema checks and fallback handling so invalid external payloads failed gracefully instead of cascading into broader process errors.
- Introduced **Redis 5.x–6.x** for short-lived caching and state coordination in selected backend flows, improving performance for repeated reads while preserving correctness through careful expiration strategy and targeted invalidation around update events.
- Worked with cloud-hosted infrastructure on **AWS** and packaged services with **Docker 20.x–23.x**, improving environment consistency and reducing friction involved in promoting backend changes across release stages.
- Strengthened backend security by refining access control checks, endpoint validation, and token-based authorization behavior, ensuring sensitive operations were protected more reliably as application complexity and user roles continued to grow.
- Supported incident diagnosis with better logging and operational traces, making it easier to isolate API failures, identify root causes faster, and reduce the amount of guesswork required when backend issues appeared during active usage.

Software Developer
ArnAmy, Inc. - Florida, United States

March 2016 to September 2018

- Developed backend services in **Node.js 10–14** for line-of-business applications, contributing to API development, data access logic, and backend workflow handling that supported operational teams with more efficient and less error-prone digital processes.
- Built maintainable **Express.js 4.x** endpoints for CRUD-heavy modules and user-driven workflows, paying close attention to validation, request sanitization, and reusable service patterns, which improved implementation consistency by **21%** across shared backend modules.
- Improved reliability of import and batch-processing routines by identifying weak points in file parsing and record validation logic, then introducing safer error handling that reduced failed import incidents by **29%** in affected operational workflows.
- Solved a persistent issue where duplicate submission events created repeated records by implementing idempotency-aware request checks and tighter server-side control of write operations, reducing cleanup effort for support and operations teams.
- Collaborated on database-backed features using **MySQL 5.7–8.x** and **PostgreSQL 10–12**, refining schema structure, query behavior, and indexing choices so that growing datasets did not degrade responsiveness of frequently used server operations.
- Helped introduce asynchronous backend patterns for notification and processing flows, allowing time-intensive operations to run more safely outside the request cycle and improving overall responsiveness for end users by **25%**.
- Contributed to early service modularization efforts that separated route handlers, business rules, and shared helpers more clearly, creating a stronger foundation for later modernization and making backend code easier to test and extend.
- Supported deployment and environment setup activities using **AWS** and **Docker 19.x–20.x**, giving the team a more consistent way to package services and reducing environment-specific issues that previously appeared during release verification.
- Enhanced error visibility by adding structured logs and clearer exception handling around fragile backend paths, which improved troubleshooting quality and shortened the time needed to reproduce and resolve recurring production defects.

- Worked closely with QA and product stakeholders to translate practical business needs into stable backend behavior, ensuring that new features were not just implemented quickly but also handled real edge cases seen in day-to-day usage.

Junior Software Developer
Centric Tech Inc. - New Jersey, United States

October 2014 to February 2016

- Supported backend development for internal and client-facing web systems by working on **Node.js 8–10** server modules, data access logic, and API-connected workflows, building an early foundation in production-minded service implementation and debugging.
- Assisted with endpoint development, input validation, and basic database operations using **JavaScript ES6+** and **Express.js 4.x**, helping improve backend implementation consistency by **18%** across smaller application modules.
- Investigated a frustrating defect where incomplete request validation caused inconsistent data to reach persistence logic, then helped tighten validation flow so invalid inputs were rejected earlier and downstream errors dropped by **23%**.
- Contributed to bug fixing across API-connected features by tracing null-related crashes, mismatched payload assumptions, and weak edge-case handling, improving stability in backend paths that had previously generated repeated QA failures.
- Helped maintain SQL-backed modules by updating queries, correcting filter logic, and supporting schema-related fixes, which improved data accuracy and reduced smaller operational issues reported by users of reporting and workflow screens.
- Participated in release support and backend verification work, building practical habits around reproducibility, log inspection, and safe change testing that later became important for more advanced service ownership responsibilities.
- Worked with senior engineers to refactor duplicated utility logic into more reusable server-side helpers, reducing inconsistency across modules and making future backend enhancements easier to implement with less repetitive code.
- Assisted with integration troubleshooting by comparing request and response behavior across environments, helping uncover configuration-related issues that caused certain backend functions to fail only after deployment.
- Supported gradual improvement of application reliability by implementing careful fixes, validating behavior against expected edge cases, and learning how disciplined debugging and maintainable backend structure reduce long-term system risk.

EDUCATION

Bachelor's degree in Computer Science
Stanford University Department of Computer Science

2010 to 2014

- Built strong foundations in **algorithms**, **databases**, and **software engineering**, applying coursework through practical team projects and production-style system design.

CERTIFICATIONS

- **Node.js Backend Engineering Professional** — 2025
Demonstrated advanced capability in building **Node.js** service layers, structuring maintainable backend modules, handling asynchronous workloads, and delivering stable API-driven systems in production environments.
- **AWS Certified Developer – Associate** — 2024
Validated practical knowledge of **AWS** application services, event-driven integrations, deployment workflows, and secure cloud-native backend delivery for scalable production systems.
- **TypeScript for Enterprise Backend Systems** — 2023
Demonstrated strong hands-on understanding of **TypeScript** service contracts, shared domain modeling, safer refactoring patterns, and runtime-risk reduction in growing backend codebases.
- **Docker & Kubernetes Foundations** — 2022
Validated working knowledge of **Docker** containerization, **Kubernetes** orchestration fundamentals, service packaging, and deployment consistency practices commonly used for modern backend platforms.
- **PostgreSQL Performance and Query Optimization** — 2021
Demonstrated practical expertise in relational modeling, index strategy, query tuning, and performance-aware backend data access patterns for operational and reporting-heavy systems.